



Ryota Takaki

Postdoctoral researcher (MPIPKS, Germany)

PERSONAL DETAILS

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RESEARCH INTERESTS

My research develops theoretical and computational frameworks for biological soft matter, focusing on how non-equilibrium processes shape material properties. I combine statistical physics, continuum mechanics, and computational modeling to study biomolecular condensates, fiber formation, and DNA organization.

PROFESSIONAL EXPERIENCE

Max Planck Institute for the Physics of Complex Systems

2021-Present

Postdoctoral researcher. Adviser: Frank Jülicher

Research: Biomolecular condensates, DNA organization, and Active matter.

EDUCATION

The University of Texas at Austin

2015-2021

Ph.D, Physics. Adviser: Dave Thirumalai

Ph.D thesis: Physical and Mathematical models for biological machines.

Kobe University

2011-2015

B.S. Earth and Planetary science. Adviser: Haruna Taichi

Research: Theoretical study on auto-catalytic chemical reaction network.

PUBLICATIONS

First/Corresponding author

- [1] **R. Takaki** and Frank Jülicher. “Fluctuation-Response Theory of Non-Equilibrium Complex Fluids”. In: *Phys. Rev. X* (Oct. 2026).
- [2] **R. Takaki**, Yahor Savich, Jan Brugués, and Frank Jülicher. “Active Loop Extrusion guides DNA-Protein Condensation”. In: *Physical Review Letters* 134.12 (2025), p. 128401.
- [3] **R. Takaki** and Dave Thirumalai. “Sequence complexity and monomer rigidity control the morphologies and aging dynamics of protein aggregates”. In: *Proceedings of the National Academy of Sciences* 121.50 (2024), e2409973121.
- [4] **R. Takaki**, Louise Jawerth, Marko Popović, and Frank Jülicher. “Theory of Rheology and Aging of Protein Condensates”. In: *PRX Life* 1 (1 Aug. 2023), p. 013006.

- [5] **R. Takaki**, Mauro L Mugnai, and D Thirumalai. “Information flow, gating, and energetics in dimeric molecular motors”. In: *Proceedings of the National Academy of Sciences* 119.46 (2022), e2208083119.
- [6] **R. Takaki**, Atreya Dey, Guang Shi, and Dave Thirumalai. “Theory and simulations of condensin mediated loop extrusion in DNA”. In: *Nature Communications* 12.1 (2021), p. 5865.
- [7] **R. Takaki**, Mauro L Mugnai, Yonathan Goldtzvik, and D Thirumalai. “How kinesin waits for ATP affects the nucleotide and load dependence of the stepping kinetics”. In: *Proceedings of the National Academy of Sciences* 116.46 (2019), pp. 23091–23099.

Co-author

- [1] Liru Feng, **R. Takaki**, Muntz Iain, Lionel Ndamb, Martine Ruer-Gruß, Frank Jülicher, and Louise Jawerth. “Fibers and Glasses are Competing Material States in Protein Condensation”. In: *Accepted (Nature Physics)* (2026).
- [2] Atreya Dey, Guang Shi, **R. Takaki**, and Dave Thirumalai. “Structural changes in chromosomes driven by multiple condensin motors during mitosis”. In: *Cell Reports* 42.4 (2023).

Under review

- [1] Dey Atreya, Shi Guang, **R. Takaki**, and Thirumalai D. “Loop Extrusion Reversal by Condensin Motor is Mediated by Catch Bonds”. In: *arXiv preprint arXiv:2605.04482* (2026).

Book chapter

- [1] Mauro L. Mugnai, **R. Takaki**, and D. Thirumalai. “Introduction to Stochastic Kinetic Models for Molecular Motors”. In: *Physics of Molecular and Cellular Processes*. Springer International Publishing, 2022, pp. 117–146.

TEACHING EXPERIENCES

Teaching assistant, The university of Texas at Austin	Spring, 2017
<i>Mathematical/computational neuroscience for undergraduate students</i>	
Teaching assistant, The university of Texas at Austin	2015-2016
<i>Physics Lab for undergraduate students</i>	

CONFERENCES/ PRESENTATIONS

Talk

Statistical Physics Youth Communications 2026	2026, Online
Soft Matter, Complex Fluids, Interfaces 2026	2026, Quy Nhon (Vietnam)
Invited research presentation	2026, Oxford University (UK)
DPG Spring Meeting	2026, Dresden (Germany)
Invited research presentation	2025, Nanyang Tech. Uni. (Singapore, online)
Physics of Life day	2023, Dresden (Germany)
Soft matter seminar at Kyushu university	2023, Fukuoka (Japan)

APS March Meeting	2023, Las Vegas (USA)
The Molecular Biology Society of Japan (Invited)	2022, Tokyo (Japan)
iPoLS district meeting	2019, Rice university (USA)
Nonlinear science research meeting	2015, Waseda university (Japan)

Poster

Driven and active amorphous matter	2025, Göttingen (Germany)
StatPhys29	2025, Florence (Italy)
Mechanics of Life II Workshop	2023, New York (USA)
7th Edwards Symposium	2023, Cambridge (UK)
Annual Biophysical Society meeting	2020, San Diego (USA)
GRC: Stochastic Physics in Biology	2019, Ventura (USA)

FELLOWSHIPS/GRANTS

Japan Society for the Promotion of Science: 6,278,000(JPY)/year	2022-2024
Shigeta Kyoiku Zaidan (Japan): 2,400,000(JPY)/year	2017-2018

AWARDS

Alexei Likhtman Poster Prize	7th Edwards Symposium (2023,UK)
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